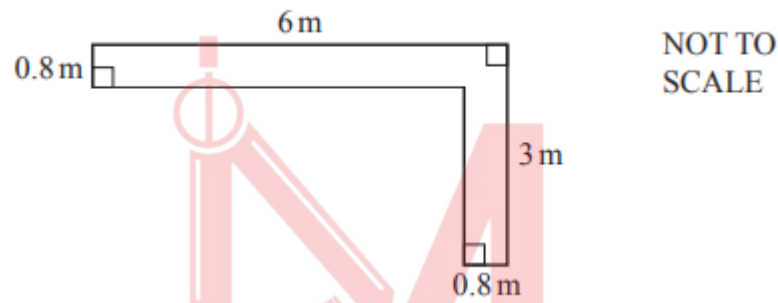


Mensuration

Worksheet 1c

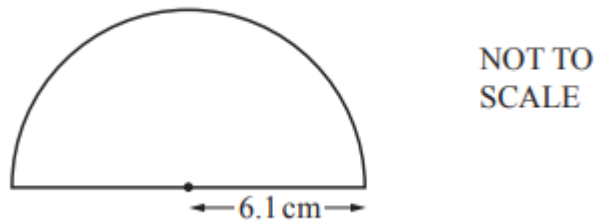
2-D Shapes Calculator Based (0580)

1. W15/qp11/q12



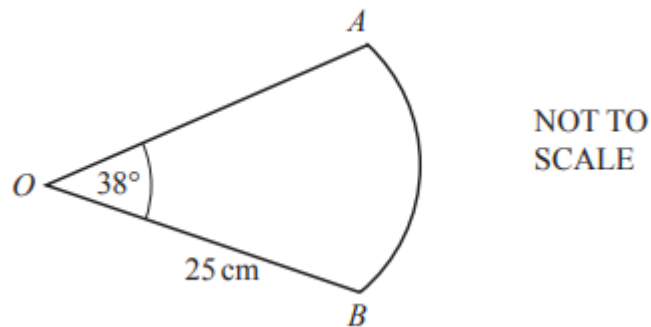
Leah is making a path in her garden using gravel. The diagram shows the path. A bag of gravel covers an area of $0.5m^2$. Work out the number of bags of gravel Leah must buy to make the path. [3]

2. W15/qp11/q18



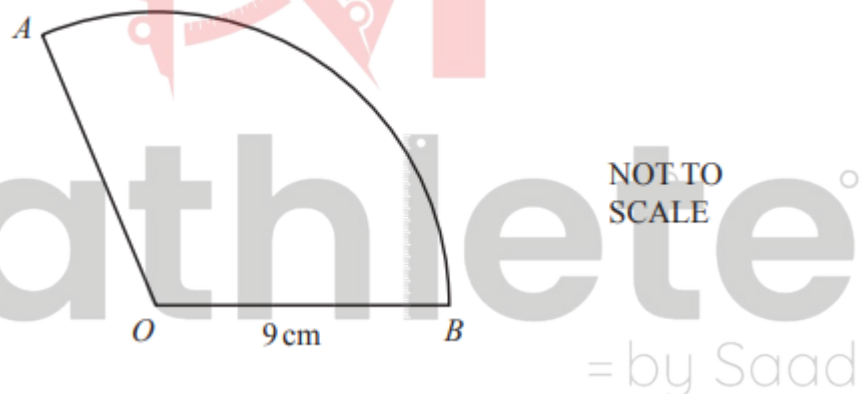
A protractor is a semi-circle of radius 6.1 cm. Calculate the perimeter of the protractor. [3]

3. M16/qp22/q11



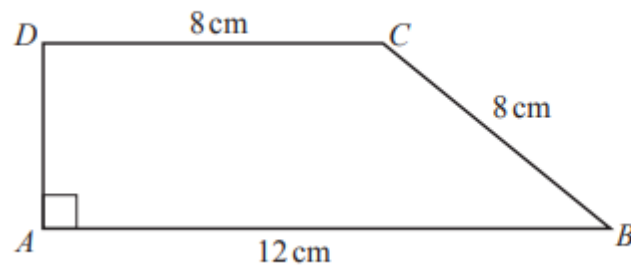
The diagram shows a sector of a circle, centre O , radius 25 cm . The sector angle is 38° . Calculate the length of the arc AB . Give your answer correct to 4 significant figures. [3]

4. S16/qp21/q20



AB is an arc of a circle, centre O , radius 9 cm . The length of the arc AB is $6\pi\text{ cm}$. The area of the sector AOB is $k\pi\text{ cm}^2$. Find the value of k . [3]

5. S16/qp21/q23

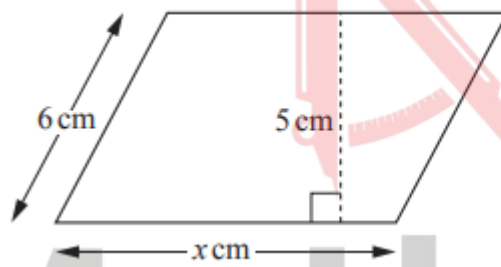


NOT TO
SCALE

Calculate the area of this trapezium.

[4]

6. S16/qp23/q3

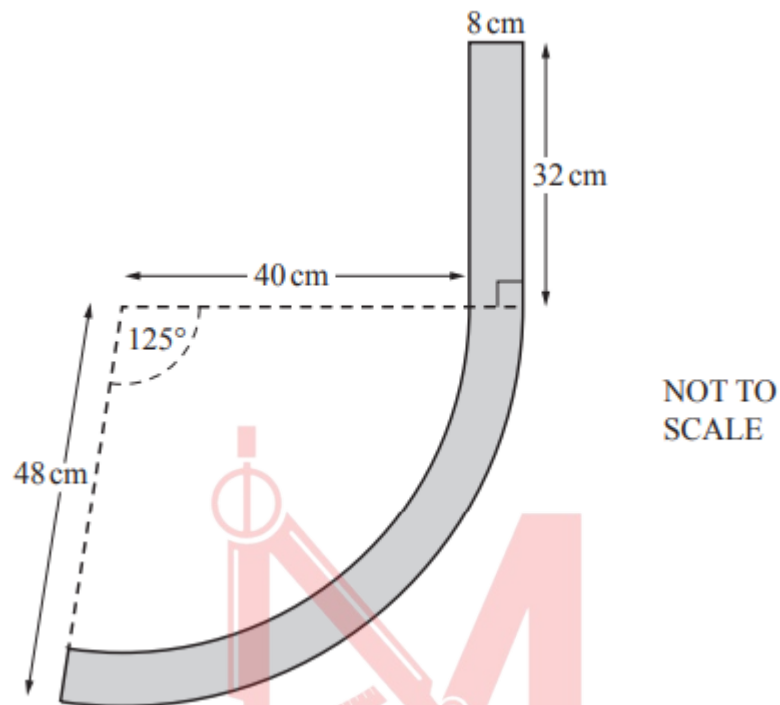


NOT TO
SCALE

The area of this parallelogram is 51.5 cm^2 . Work out the value of x . [2]

Mathlete[®]
= by Saad

7. W16/qp22/q17

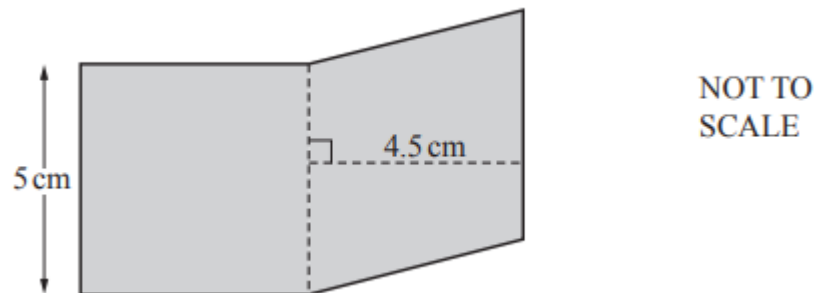


The diagram shows the cross section of part of a park bench. It is made from a rectangle of length 32 cm and width 8 cm and a curved section. The curved section is made from two concentric arcs with sector angle 125° . The inner arc has radius 40 cm and the outer arc has radius 48 cm. Calculate the area of the cross section correct to the nearest square centimetre. [5]

= by Saad

8. W16/qp23/q14

The shaded shape is made by joining a square and a rhombus.



Work out

- the perimeter of the shaded shape. [1]
- the area of the shaded shape. [2]

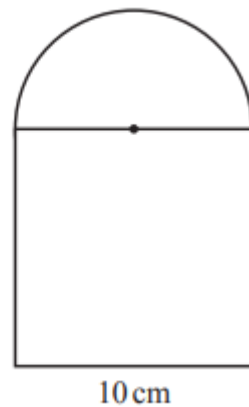
9. W17/qp22/q23



The diagram shows a sector of a circle, centre O and radius 6 cm. The sector angle is 30° . The area of the shaded segment is $(kr - c) \text{ cm}^2$, where k and c are integers.

Find the value of k and the value of c . [3]

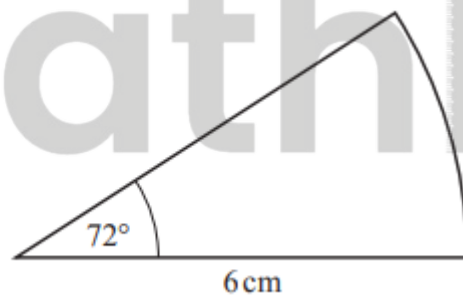
10. W17/qp23/q20



NOT TO
SCALE

The diagram shows a shape made from a square and a semi-circle. Calculate the area of the shape. Give the units of your answer. [5]

11. W18/qp23/q19



NOT TO
SCALE

= by Saad

The diagram shows a sector of a circle with radius 6cm and sector angle 72° . The perimeter of this sector is $(p + q\pi)cm$. Find the value of p and the value of q . [3]

Answer Key

1. W15/qp11/q12

12	14 nfw	3	M2 for $(0.8 \times 6 + 2.2 \times 0.8) \text{ oe} \div 0.5 \text{ oe}$ or M1 for $0.8 \times 6 + 2.2 \times 0.8 \text{ oe}$ If zero scored, SC1 for <i>their</i> attempt at $\text{area} \div 0.5$ and SC1 for any non-integer answer for <i>their</i> value $\div 0.5$ rounded up
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2. W15/qp11/q18

18	31.4 or 31.36 to 31.37	3	M2 for $\left[\frac{2}{2} \times\right] 6.1 \times \pi + 2 \times 6.1 \text{ oe}$ or B2 for 19.16 to 19.17 or 19.2 or M1 for $6.1 \times \pi$ or for $12.2 \times \pi$
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3. M16/qp22/q11

11	16.58 cao	3	B2 for 16.6 or 16.580 to 16.583 final answer or 16.58 not as final answer or M1 for $\frac{38}{360} \times 2 \times \pi \times 25$ and B1 for rounding their more accurate answer correctly to 4sf
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4. S16/qp21/q20

20	27	3	M2 for $\frac{6\pi}{\pi \times 2 \times 9} \times \pi \times 9^2 \text{ oe}$ or M1 for $\frac{6\pi}{\pi \times 2 \times 9} \text{ oe}$
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5. S16/qp21/q23

23	69.3 or 69.28...	4	M2 for height = $\sqrt{8^2 - 4^2}$ or M1 for $4^2 + h^2 = 8^2 \text{ oe}$ and M1 for $\frac{1}{2}(8+12) \times \text{their perp height} \text{ oe}$
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6. S16/qp23/q3

3	10.3 oe	2	M1 for $5x = 51.5$ oe
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7. W16/qp22/q17

17	1024 cao	5	B4 for 1023 to 1024.0... or 1020 or M3 for $\frac{125}{360} \times \pi \times 48^2 - \frac{125}{360} \times \pi \times 40^2 + 32 \times 8$ or M1 for $\frac{125}{360} \times \pi \times 48^2$ or $\frac{125}{360} \times \pi \times 40^2$ and M1 for $32 \times 8 + k\pi$ If B0 scored B1 for <i>their</i> more accurate decimal answer rounded correctly to an integer
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8. W16/qp23/q14

14 (a)	30	1	
(b)	47.5	2	M1 for 4.5×5 oe

9. W17/qp22/q23

23	$[k =] 3$ $[c =] 9$	3	M1 for $\frac{30}{360} \times \pi \times 6^2$ M1 for $\frac{1}{2} \times 6 \times 6 \times \sin 30$
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10. W17/qp23/q20

20	139 or 139.2 to 139.3	4	M3 for $10^2 + \frac{1}{2} \times \pi \times 5^2$ or M2 for $\frac{1}{2} \times \pi \times 5^2$ or M1 for radius = 5 or [area of square]10^2
	cm ²	1	

11. W18/qp23/q19

19	$[p =] 12$ $[q =] \frac{12}{5}$ oe	3	B1 for $[p =] 12$ and B2 for $[q =] \frac{12}{5}$ or M1 for $\frac{72}{360} [\times \pi] \times 2 \times 6$ oe
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